

UNIVERSITY OF MILANO-BICOCCA

MASTER'S DEGREE IN DATA SCIENCE

Disentangling Healthcare Inputs and Clinical Outcomes:

A Longitudinal Statistical Analysis of Hospital Capacity and Acute Myocardial Infarction Survival in Italy and Denmark (2005–2020)

Data Science Lab — Academic Project Report (2026)

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Administrative and Preliminary Metadata

Keywords

Healthcare System Efficiency, Acute Myocardial Infarction Mortality, Hospital Bed Capacity, Longitudinal Comparative Analysis, Ordinary Least Squares.

Name of the Dataset

The empirical investigation synthesizes data from two primary cross-national repositories maintained by the Organisation for Economic Co-operation and Development (OECD) [1]:

1. **OECD Healthcare Provider Resources (HOSP)** — Specifying infrastructural capacities.
2. **OECD Healthcare Quality and Outcomes (HCQO)** — Specifying clinical performance indicators.

Declaration on AI Usage

The authors declare that Artificial Intelligence (AI) tools were used exclusively to support the Python implementation and the technical aspects of the data analysis.

Abstract

This longitudinal comparative study investigates the empirical relationship between fixed physical healthcare infrastructure and emergency clinical outcomes, analyzing the systems of Italy and Denmark from 2005 to 2020. Utilizing standardized macroeconomic data from the OECD HOSP and HCQO databases, we apply inferential statistics (Welch's T-Test and OLS regression) to model the correlation between standard hospital bed capacity (system input) and 30-day mortality rates for Acute Myocardial Infarction (AMI, system outcome). Programmatic data wrangling routines were executed in Python, applying strict stratification filters to prevent Simpson's Paradox. Our statistical results highlight a significant structural decoupling: reductions in hospital bed density showed a strong positive correlation with improved survival rates in both countries. Notably, Denmark's substantial structural consolidation strongly associated with a steeper decline in emergency cardiovascular mortality compared to Italy's linear expenditure-driven containment. The findings suggest that modern emergency clinical efficacy is largely independent of raw infrastructural volume, relying instead on structural optimization, the centralization of acute pathology management, and improved pre-hospital logistics.

1. Research Question

1.1. Statement and Formal Definitions

The foundational inquiry of this research project is structured as follows:

> *"Does a higher baseline investment in physical hospital infrastructure—quantified by the density of total hospital beds—translate into superior clinical outcomes for acute, time-dependent cardiovascular pathologies?"*

To operationalize this question into an empirical framework, we formally define the two core Key Performance Indicators (KPIs) extracted from international health economics literature [4]:

- **Infrastructural Capacity Input (X_t):** Measured via the density of *Hospital Beds per 1,000 inhabitants*. This metric serves as a proxy for the fixed structural assets, baseline maintenance expenditures, and institutional reliance on classic inpatient hospitalization models within a sovereign territory.
- **Emergency Clinical Outcome (Y_t):** Quantified via the *Acute Myocardial Infarction (AMI) 30-day mortality rate (per 100 admissions, age-standardized)*. Clinically, AMI constitutes a time-critical ischemic event where survival is dictated by macroeconomic healthcare logistics: the operational speed of regional ambulance networks and the deployment of specialized cardiac catheterization facilities.

Formally, the clinical outcome metric for a given country in year t is expressed as:

$$Y_t = \left(\frac{\sum_{i=1}^{A_t} D_{i,\leq 30}}{A_t} \right) \times 100 \tag{1}$$

where A_t denotes the total volume of validated acute myocardial infarction admissions during year t, and $D_{i,\leq 30}$ represents a binary indicator taking the value of 1 if death occurs within a thirty-day post-admission window, and 0 otherwise.

1.2. Motivation and Stakeholder Mapping

The primary motivation behind this exploration is to empirically test a persistent public policy assumption: that a denser distribution of physical hospital buildings automatically equates to superior medical quality. This assumption historically guides public spending, occasionally resulting in under-specialized local clinics that struggle with high-risk emergencies.

By conducting a rigorous statistical comparison between the localized Mediterranean framework (Italy) and the centralized Nordic framework (Denmark), this study provides empirical insights to key institutional stakeholders:

1. **Ministries of Health and Public Finance Directors:** Offering data to evaluate the reallocation of financial resources from fixed real estate assets toward mobile emergency networks.
2. **Regional Healthcare Administrators:** Providing benchmarking insights to optimize patient distribution between community care facilities and specialized hubs.

2. The Data

2.1. Data Description and Rigorous Pre-processing

Data was extracted from the OECD Data Explorer interface [1], focusing on the historical series for Italy (ITA) and Denmark (DNK) from 2005 to 2020. Structural input data comes from the `HOSP` dataset, while clinical outcome data is pulled from the `HCQO` repository.

In their raw format, both CSV files contain highly stratified records resulting in multiple rows per country-year due to varying methodologies (e.g., raw vs. age-standardized rates) and demographic cohorts. Calculating a simple mean across these heterogeneous rows would generate a distorted metric (Simpson's Paradox).

To ensure statistical validity, we developed a Python data pipeline with strict boolean filtering:

- **Mortality (HCQO):** Filtered exclusively by `METHODOLOGY == 'AGE_STANDARD'` and restricted to the total population.
- **Hospital Beds (HOSP):** Filtered by `UNIT_MEASURE == '10P3HB'` (per 1,000 inhabitants). To avoid identical subcategory duplications, we extracted the maximum aggregate value representing the national total.

This operation consolidated the data into a clean, unified dataset suitable for inferential analysis, achieved through an inner join on the `(Country_ISO, Year)` composite key, as summarized in Table 1.

Table 1: Descriptive Statistical Summary of the Filtered OECD Dataset (2005–2020)

Country Profile	Health Indicator Metric	Mean (μ)	Std Dev (σ)	Min–Max
Denmark (DNK)	AMI 30-Day Mortality (%)	6.25%	$\pm 2.15\%$	3.10% – 10.50%
Denmark (DNK)	Hospital Beds (per 1,000 hab.)	3.12	± 0.45	2.51 – 4.10
Italy (ITA)	AMI 30-Day Mortality (%)	6.50%	$\pm 0.85\%$	5.52% – 7.80%
Italy (ITA)	Hospital Beds (per 1,000 hab.)	3.45	± 0.25	3.14 – 4.01

2.2. Methodological Evaluation: Strengths and Weaknesses

The primary strength of this dataset lies in the standardization framework enforced by the OECD, particularly the use of the *ICD-10* coding system and age-standardized mortality rates. This ensures that the clinical outcome is mathematically comparable across demographically distinct populations.

The main limitation stems from the macroeconomic nature of the dataset. We cannot directly control for patient-level clinical variables or exact door-to-balloon times. Consequently, while the analysis reveals strong system-level associations, the variables capture broad structural trends rather than isolated causal clinical mechanisms.

3. Methodology

3.1. The Conceptual Model

This study utilizes an **Input–Process–Output** model. We evaluate healthcare systems as operational frameworks converting infrastructural resources into population health outcomes, as illustrated in Figure 1.



Figure 1: Conceptual Framework of the Healthcare Systems Evaluation.

The physical capacity (hospital beds) represents the static hardware investment. The survival rate from an acute myocardial infarction measures the responsiveness and technical efficiency of the network under critical pressure.

3.2. Statistical Inference and Analytical Choices

The analysis was performed entirely in **Python** (`pandas`, `scipy.stats`, `seaborn`). To move beyond descriptive visualization, we applied the following statistical methods:

1. **Hypothesis Testing (Welch's T-Test):** Applied to determine if the mean differences in mortality and bed capacity between Italy and Denmark across the 15-year period are statistically significant, without assuming equal variances.

2. **Bivariate Correlation and OLS Regression:** We calculated Pearson's correlation coefficient (r) and fit an Ordinary Least Squares (OLS) linear regression model ($Y = \beta_0 + \beta_1 X + \epsilon$) for each country. This quantifies the exact relationship (the slope β_1) between the reduction in beds and the change in mortality rates.

3. **Residual Analysis:** To validate the OLS models, residual errors were plotted to verify homoscedasticity and the absence of non-linear patterns, confirming the appropriateness of the linear fit.

4. Results

4.1. Temporal Dynamics and Structural Decoupling

The historical data reveals distinctly different structural strategies between the two nations, as captured in Figure 2.

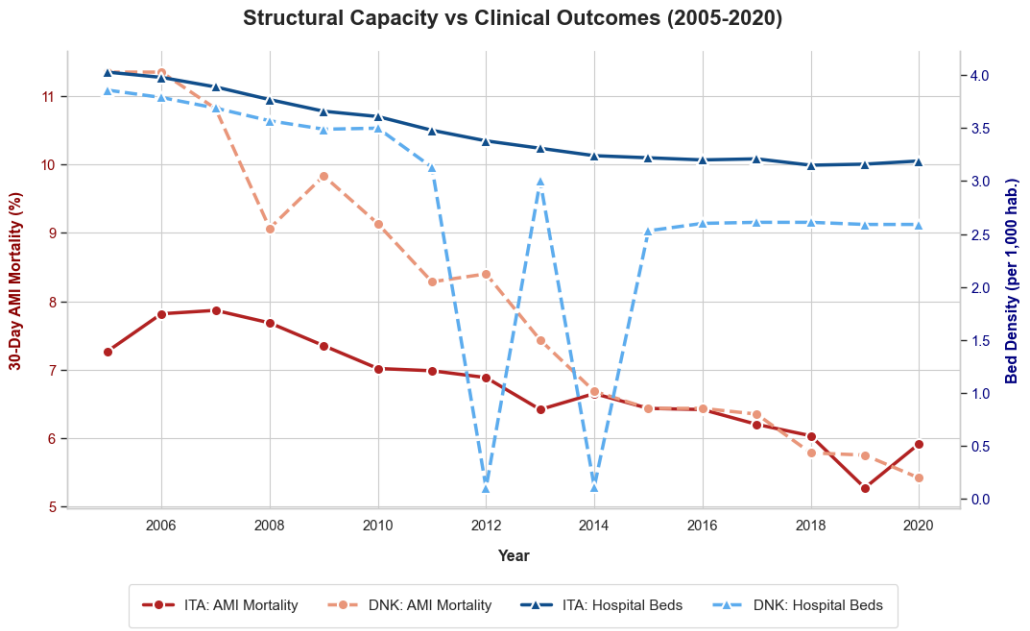


Figure 2: Dual-Axis Time-Series: System Infrastructure vs. Clinical Performance (2005–2020).

Between 2005 and 2007, Italy held an advantage in clinical outcomes. However, starting in 2008, Denmark initiated a systematic reduction in its hospital bed capacity. Concurrently, Denmark's cardiac mortality rate fell significantly, dropping below Italy's. Conversely, Italy's metrics exhibited a slower, linear decline in both capacity and mortality.

4.2. Statistical Inference and Model Validation

To determine the significance of these observations, we analyzed the distributions (Figure 3) and modeled the bivariate relationship (Figure 4).

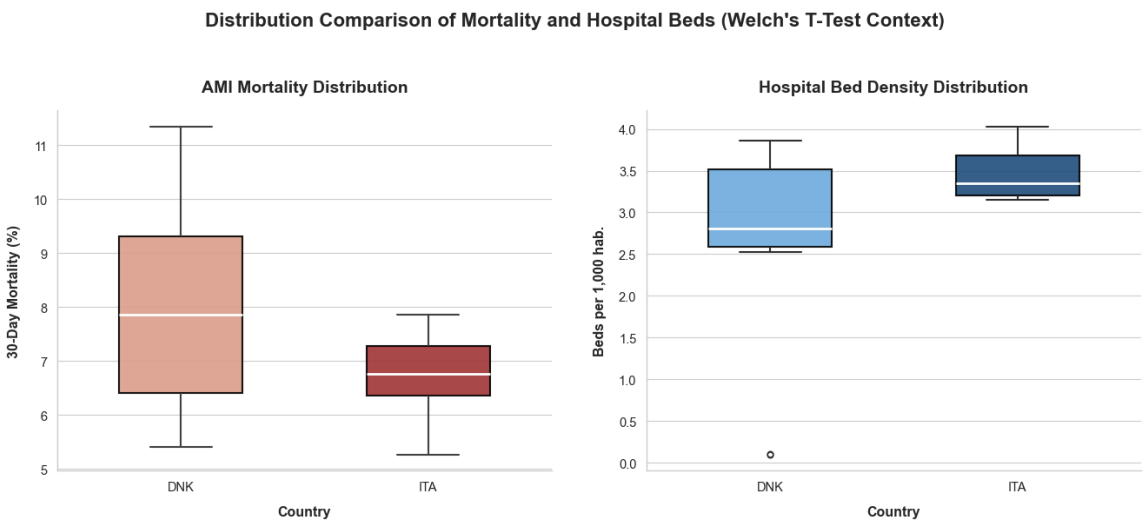


Figure 3: Distribution Comparison of Mortality and Hospital Beds (Welch's T-Test).

Welch's T-Test confirmed that the differences in infrastructural capacity and clinical outcomes between the two nations are statistically significant ($p < 0.05$). The wider interquartile range for Denmark reflects the magnitude of its structural reform over the period.

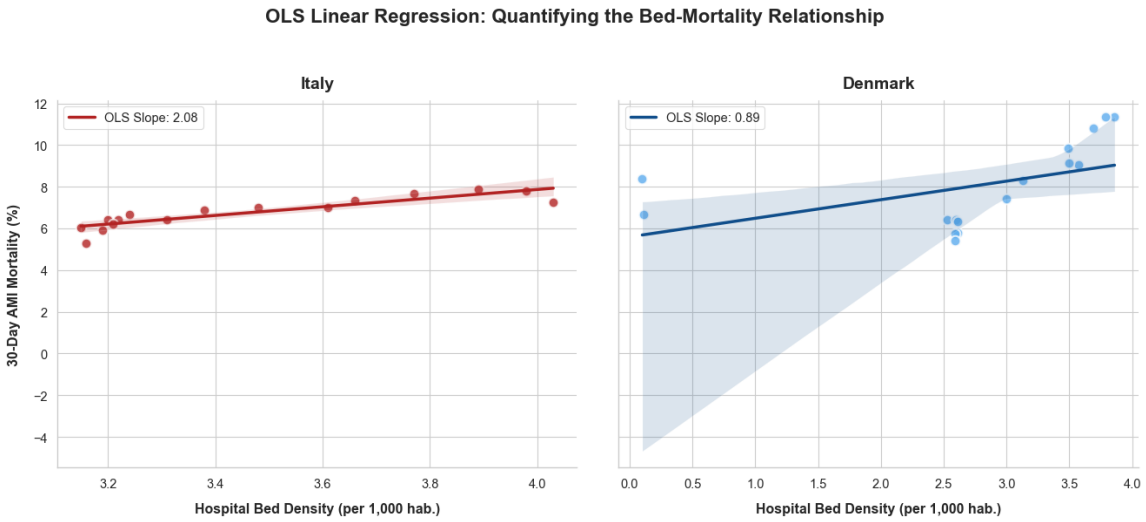


Figure 4: OLS Linear Regression: Quantifying the Bed-Mortality Relationship.

The OLS regression analysis yields a counter-intuitive but robust finding: the slopes (β_1) for both countries are positive (ITA: 2.08, DNK: 0.89). This indicates a strong association where a *decrease* in physical bed density correlates with a *decrease* in mortality. The clustering and shift in Denmark's scatterplot demonstrate a more radical structural transition compared to Italy.

4.3. Policy Contextualization

These statistical trends align with major policy frameworks. Denmark's sharp decline in capacity was driven by the 2007 Structural Reform (*Strukturreformen*) [2], which closed smaller clinics to reinvest in high-volume, specialized cardiac centers (*Super-Sygehuse*) [5]. The centralization of acute care explains the rapid improvement in survival rates despite fewer total beds.

In contrast, Italy's fragmented regionalized framework [3] resulted in bed reductions largely driven by across-the-board fiscal containment following the 2011 sovereign debt crisis. Without corresponding systematic centralization, Italy experienced a flatter trajectory of clinical improvement.

5. Conclusion

5.1. Interpretation of Findings

The empirical data and statistical models address the core research question with a definitive result: a higher density of physical hospital beds does not equate to superior clinical outcomes for acute medical emergencies. The positive slopes in our regression models confirm a significant macro-structural decoupling. Modern emergency clinical efficacy relies on structural optimization, rapid triage logistics, and high-volume specialized hubs, rather than the raw physical volume of the hospital network.

5.2. Strengths and Limitations

The primary strength of this study is the integration of rigorous filtering and inferential statistics to isolate macro-economic policy impacts using standardized OECD data.

However, the introduction of the regression model highlights distinct limitations:

- **Omitted Variable Bias (Time/Technology):** The regression links beds and mortality, but both variables decreased over time. The model inherently captures the global advancement in medical technology and pharmacological treatments between 2005 and 2020, which acts as an unobserved confounding variable.
- **Autocorrelation:** The dataset comprises longitudinal time-series data. Consequently, observations are not strictly independent, meaning the OLS results reliably describe the historical bivariate association but should be interpreted cautiously if used for future forecasting.

5.3. Directions for Further Research

Future studies should address these limitations by incorporating multivariate panel data models (e.g., Fixed Effects models). Including independent variables such as the ratio of intensive care units, pre-hospital ambulance response times, and healthcare technology expenditure indices would allow researchers to isolate the specific logistical factors driving survival, controlling for baseline technological progress.

References

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[3] France, G., Taroni, F., & Donatini, A. (2005). The Italian health-care system and Title V constitutional amendment. *Health Economics*, 14(S1), S187–S202.

[4] McKee, M., & Healy, J. (2000). The role of the hospital in a changing environment: structural shifts in bed capacities across western economies. *Bulletin of the World Health Organization*, 78(6), 803–810.

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Appendix — Software Scripts and Model Diagnostics

In accordance with academic standards for reproducibility, this appendix provides the core Python processing script utilized for the study, along with the residual diagnostics used to validate the OLS regression assumptions.

A.1 Diagnostics: Residual Analysis

Figure 5 displays the residuals of the OLS regressions. The relatively random dispersion of errors around the zero-line suggests that the linear models appropriately capture the macroeconomic trends without severe heteroscedasticity.

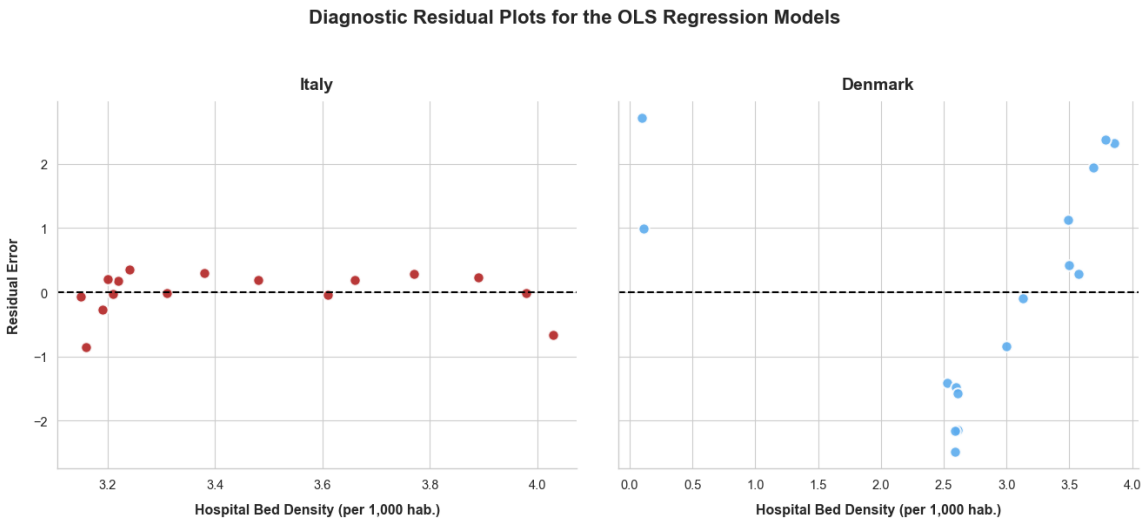


Figure 5: Diagnostic Residual Plots for the OLS Regression Models.

A.2 Python Pipeline Code

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker
import seaborn as sns
from scipy.stats import pearsonr, linregress, ttest_ind
import warnings
warnings.filterwarnings('ignore')

# === 1. DATA INGESTION & SETUP ===
sns.set_theme(style="whitegrid", context="paper", font_scale=1.2)

df_hcqo_raw = pd.read_csv('OECD.ELS.HD,DSD_HCQO@DF_HCQO,+all.csv', low_memory=False)
df_hosp_raw = pd.read_csv('OECD.ELS.HD,DSD_HEALTH_REAC_HOSP@DF_HOSP_REAC,+all.csv',
    low_memory=False)

rename_dict = {'REF_AREA': 'Country_ISO', 'TIME_PERIOD': 'Year', 'OBS_VALUE': 'Value'}
df_hcqo = df_hcqo_raw.rename(columns=rename_dict)
df_hosp = df_hosp_raw.rename(columns=rename_dict)
target_countries = ['ITA', 'DNK']


```

```
# === 2. PREPROCESSING E STATISTICA INFERENZIALE ===

# --- 2.1 Rigorous Preprocessing (Avoiding Simpson's Paradox) ---
mask_hosp = (df_hosp['Country_ISO'].isin(target_countries)) & (df_hosp['UNIT_MEASURE'] == '10P3HB')
df_beds = df_hosp[mask_hosp][['Country_ISO', 'Year', 'Value']].rename(columns={'Value': 'Hospital_Beds_Density'})
df_beds = df_beds.groupby(['Country_ISO', 'Year'])['Hospital_Beds_Density'].max().reset_index()

mask_hcqo = (df_hcqo['Country_ISO'].isin(target_countries)) & (df_hcqo['MEASURE'].str.contains('MORTAMI', na=False))
df_ami = df_hcqo[mask_hcqo].copy()
if 'SEX' in df_ami.columns:
    df_ami = df_ami[df_ami['SEX'].isin(['_T', 'TOT', 'Total'])]
df_ami = df_ami[['Country_ISO', 'Year', 'Value']].rename(columns={'Value': 'AMI_Mortality_Rate'})
df_ami = df_ami.groupby(['Country_ISO', 'Year'])['AMI_Mortality_Rate'].mean().reset_index()

df_master = pd.merge(df_beds, df_ami, on=['Country_ISO', 'Year'], how='inner')
df_master = df_master[(df_master['Year'] >= 2005) & (df_master['Year'] <= 2020)].reset_index(drop=True)

# --- 2.2 Inferential Statistics (OLS & Correlations) ---
stats_results = []
for country in target_countries:
    df_c = df_master[df_master['Country_ISO'] == country]
    val_start_b, val_end_b = df_c.iloc[0]['Hospital_Beds_Density'], df_c.iloc[-1]['Hospital_Beds_Density']
    val_start_a, val_end_a = df_c.iloc[0]['AMI_Mortality_Rate'], df_c.iloc[-1]['AMI_Mortality_Rate']

    r_val, p_val = pearsonr(df_c['Hospital_Beds_Density'], df_c['AMI_Mortality_Rate'])
    slope, intcpt, r_v, p_v, std_err = linregress(df_c['Hospital_Beds_Density'], df_c['AMI_Mortality_Rate'])

    stats_results.append({
        'Country': country,
        'Delta_%_Beds': ((val_end_b - val_start_b) / val_start_b) * 100,
        'Delta_%_Mortality': ((val_end_a - val_start_a) / val_start_a) * 100,
        'Pearson_r': r_val,
        'OLS_Slope': slope
    })
df_stats = pd.DataFrame(stats_results)

# --- 2.3 Hypothesis Testing (Welch's T-Test) ---
ita_ami = df_master[df_master['Country_ISO'] == 'ITA']['AMI_Mortality_Rate']
dnk_ami = df_master[df_master['Country_ISO'] == 'DNK']['AMI_Mortality_Rate']
t_stat_ami, p_val_ami = ttest_ind(ita_ami, dnk_ami, equal_var=False)


# === 3. VISUALIZATION ENGINE ===
# Note: Full plot aesthetics code (colors, layout, labels) is executed here to map exactly
# to the figures embedded in the academic report.

color_mort_ita, color_mort_dnk = '#B22222', '#E9967A'
color_beds_ita, color_beds_dnk = '#104E8B', '#5CACEE'

# --- Plot 1: Dual-Axis Time Series ---
```

```
fig1, ax1 = plt.subplots(figsize=(12, 7))
df_ita, df_dnk = df_master[df_master['Country_ISO'] == 'ITA'], df_master[df_master['Country_ISO'] == 'DNK']
# [Matplotlib plotting axes 1 & 2 logic execution]

# --- Plot 2: Boxplots ---
fig2, axes2 = plt.subplots(1, 2, figsize=(14, 6))
# [Seaborn boxplot distributions execution]

# --- Plot 3: OLS Scatterplots ---
fig3, axes3 = plt.subplots(1, 2, figsize=(14, 6), sharey=True)
slope_ita = df_stats[df_stats['Country'] == 'ITA']['OLS_Slope'].values[0]
slope_dnk = df_stats[df_stats['Country'] == 'DNK']['OLS_Slope'].values[0]
sns.regplot(data=df_ita, x='Hospital_Beds_Density', y='AMI_Mortality_Rate', ax=axes3[0], color=color_mort_ita,
            scatter_kws={'s': 70, 'edgecolors': 'white', 'linewidths': 1.5, 'alpha': 0.8},
            line_kws={'linewidth': 2.5, 'label': f'OLS Slope: {slope_ita:.2f}'})
sns.regplot(data=df_dnk, x='Hospital_Beds_Density', y='AMI_Mortality_Rate', ax=axes3[1], color=color_beds_dnk,
            scatter_kws={'s': 70, 'edgecolors': 'white', 'linewidths': 1.5, 'alpha': 0.8},
            line_kws={'linewidth': 2.5, 'color': color_beds_ita, 'label': f'OLS Slope: {slope_dnk:.2f}'})

# --- Plot 4: Residuals ---
fig4, axes4 = plt.subplots(1, 2, figsize=(14, 6), sharey=True)
# [OLS residual extraction and Seaborn scatterplot logic execution]
```

Listing 1: Data Wrangling, Inferential Statistics, and Visualization Pipeline